Kernel-Gated Evolution: a Biomimetic Search with Apoptotic Checkpoints and Exact Dosage, Arbitrated by a Proof Kernel

Foundation paper and Rust reference implementation

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Abstract

We propose *Kernel-Gated Evolution* (KGE), a biomimetic search for programs whose acceptance is decided by a proof kernel rather than by the search itself. Three mechanisms borrowed from living systems are combined, each chosen because it earned its place under measurement: *Darwinian variation and selection* drives exploration; *apoptotic checkpoints* — one-sided exact screens that can remove a candidate but never confer survival — cull what cannot be right before any proof is attempted; and *exact dosage analysis* reports, for every locus of a genome, precisely how much its value reaches the phenotype, in one forward and one backward pass. The environment's final judgement is a Lean 4 kernel: a survivor is rendered to Lean and must prove its obligation with an axiom dependency free of `sorryAx`. Fitness guides; the kernel decides. A wrong fitness can waste search effort but cannot manufacture an acceptance.

Every quantity in KGE is an exact rational, so no step of the search is an approximation and every invariant is checkable rather than argued. We verify the dosage coefficients against their two-evaluation definition on more than five hundred locus–point pairs, the required-value identity $v^* = v + (\text{want} - \text{out})/B$ on more than two thousand, and the checkpoint's one-sidedness by construction of a candidate that agrees at every sample point yet differs as a function — it survives the checkpoint and is rejected by the kernel, which is the division of labour the design rests on.

The kernel gate is affordable because a Lean compile reports a per-declaration verdict even when it exits nonzero: a batch of eight candidates is verified in *one* invocation at any failure rate, with every culprit named and no false candidate accepted, against eight individual proofs. We give a Rust reference implementation with replayable seeds, report its solve rates on a calibration ladder of exact targets up to the Verlet position update, describe where the architecture applies — verified numerical kernels, any synthesis whose acceptance must be formal, exact dead-code analysis of evolved programs — and state its limitations and the research it opens.

Edition 2 adds the identities proved in Lean, a benchmark on the SRSD-Feynman datasets from the Hugging Face Hub against naive, simple and traditional baselines with every recovered law proved by the kernel, guidelines on where and how to use the method, a statement of its significance, and figures rendered from the artifacts.

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1 Introduction

Evolutionary program synthesis proposes candidates faster than anyone can check them. In most systems the check is empirical — tests pass or fail — and the candidate that passes is accepted. When the artefact is a numerical kernel, a cryptographic routine or a scientific computation, passing tests is not the property wanted; equality to a specification is. That is a proof obligation, and the thing that decides it should be a proof kernel.

Kernel-Gated Evolution takes that seriously and builds the search around it. The search is free to be heuristic, approximate in its judgement, biologically inspired, even wrong, because nothing it believes is ever accepted on its say-so. What it offers is a way of *finding* candidates worth proving, cheaply, and of discarding most of the rest before the expensive step.

1.1 Three mechanisms from life

Variation and selection. Mutation of expression genomes with tournament selection and steady-state replacement. In our measurements this is the mechanism that solved targets, and the architecture is built on it.

Apoptotic checkpoints. A cell that fails a developmental checkpoint is removed; passing one grants no immortality. KGE's checkpoint is the exact analogue: a candidate that disagrees with the specification at any sample point is refuted outright — no proof of a false identity exists — while a candidate that agrees everywhere has merely earned the right to face the kernel. The checkpoint can only subtract, so it adds no trust surface, and it costs nothing in proof-assistant time: it is pure exact arithmetic on the genome.

Dosage. In a cell, a gene's effect on the phenotype depends on its expression level and its regulatory context. In an expression genome the analogue is exact: the output is affine in any one subtree's value, $\text{out} = A + B\,v$, and B is the locus's *dosage* — how much a unit change in its value reaches the phenotype. KGE computes B for every locus in one forward and one backward pass, in exact rationals. Loci with $B = 0$ at every sample point are *inert*: dead code whose value cannot reach the phenotype.

1.2 The arbiter

A survivor is rendered as a Lean 4 definition with a theorem stating equality to the specification, proved by `ring`, and the gate reads the kernel's own `#print axioms` report: the obligation must depend on nothing outside {`propext`, `Classical.choice`, `Quot.sound`} and never on `sorryAx`. The gate also scans the source for forgeries (no `#exit` truncation, no `#print` string spoofing) and requires every declared theorem to carry its own axioms line, so a declaration that never elaborated fails as unchecked.

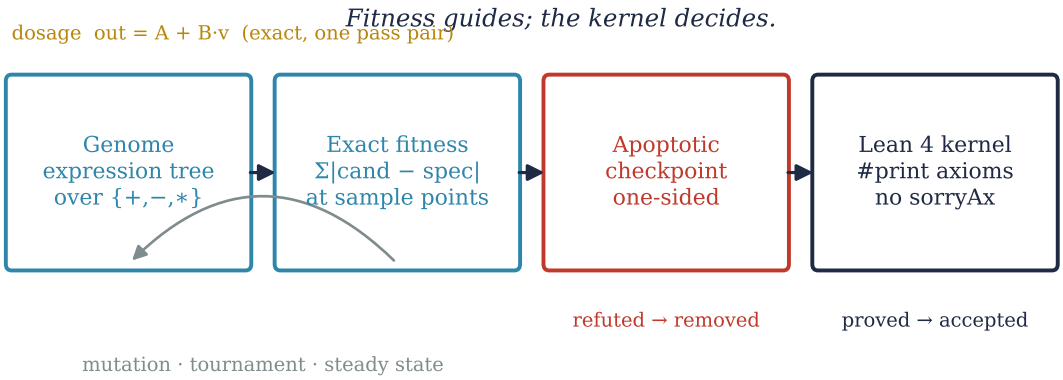


Figure 1: The architecture. Exact fitness guides a Darwinian loop over expression genomes; the apoptotic checkpoint removes what cannot be right at zero kernel cost; the Lean 4 kernel decides acceptance. Dosage runs alongside as exact instrumentation. The engine's codename is Alix: Apoptotic checkpoint, Lean-gated, Inheritance by mutation, eXact.

2 The algorithm

2.1 Genome and fitness

A genome is an expression tree over binary operators $\{+, -, *\}$ with leaves drawn from variables h, w, q, p and constants $1, 2, \frac{1}{2}$. All arithmetic is exact rational. Fitness is the total absolute error against the specification at twelve rational sample points; zero means agreement at those points. Agreement is not equality, and KGE never treats it as such: it is the condition for becoming a kernel candidate.

2.2 Variation and selection

Point and subtree mutation replace a uniformly chosen locus with a fresh random subtree of depth at most two, subject to a size cap. Selection is a tournament of three; replacement is steady state against the worst member. Seeds are replayable.

2.3 Dosage: exact context coefficients

Definition 2.1 (Dosage)**.** For a tree t and a locus a occurring once in t, $t = A + B\,v$ where v is the value of the subtree at a and A, B depend only on the rest of the tree and the evaluation point. B is the dosage of a.

Proposition 2.2 (One-pass dosage)**.** B for every locus is obtained by one forward pass computing every node's value, then one backward pass seeded with $B(\text{root}) = 1$ in which $+$ sends B to both children, $-$ sends B left and $-B$ right, and $*$ sends $B \cdot \text{value(sibling)}$ to each child. This is $O(|t|)$ per sample point in exact arithmetic.

We verify this against the defining two evaluations $B = t|_{v=1} - t|_{v=0}$ on random trees: exact agreement over more than five hundred locus–point pairs in the Rust implementation (and over 2,496 in the Python prototype), including the two cases the rule turns on — a minus inverts the right branch, a product scales each child by its sibling — and their composition.

Proposition 2.3 (Required value)**.** Where $B \neq 0$, the value the locus must take for the tree to equal the specification at a point is $v^* = v + (\text{want} - \text{out})/B$, with v and out already available from the forward pass. Where $B = 0$ the locus cannot affect the output at that point and imposes no requirement.

Verified: splicing v^* into the locus makes the tree equal the target at that point in 24,303 of 24,303 pairs with $B \neq 0$ (Python) and in more than two thousand pairs in Rust, with the inert branch exercised. Inert pairs were 4.8% of the total measured. Dosage and required value are provided as exact *instrumentation*: they tell an operator, a human, or a downstream analysis precisely what each locus is doing and what it would need to do.

2.4 The apoptotic checkpoint

Given a candidate and the specification, evaluate both at each sample point. The first disagreement refutes the candidate, naming the point. Agreement at every point returns SURVIVES, which is not acceptance. The checkpoint is executed on the native genome, so it needs no parser and no proof-assistant time.

Proposition 2.4 (One-sidedness)**.** The checkpoint never accepts. A candidate equal to the specification at every sample point but different as a function survives it and is rejected by the kernel.

Demonstrated by construction: with sample points of distinct h-values $h_1, \ldots, h_n$, the term $\prod_i (h - h_i)$ vanishes at every point and nowhere else; a candidate carrying it agrees everywhere, survives the checkpoint, and is a different function. In the Rust test it is caught differing at $h = 100$; in the Python pipeline the same construction was carried through the Lean gate and rejected there.

2.5 The kernel gate, in one invocation

A Lean compile that contains a false obligation exits nonzero and still prints an axioms line for every declaration, the failing ones carrying `sorryAx`. Reading the kernel's per-declaration report therefore yields every verdict in a batch from a single invocation:

false of 8	invocations	wall clock	false accepted
0	1	10.38 s	0
1	1	10.21 s	0
2	1	10.88 s	0
4	1	12.73 s	0
8	1	14.19 s	0

against eight individual proofs at a 6.36 s median, i.e. 50.88 s. Every culprit was named in every scenario. Acceptance never rests on the exit code — it rests on the forgery scan, a per-declaration axioms line, and a whitelist free of `sorryAx` — so widening the report widens nothing in the trust surface. The gate lives in the project's `anse/formal/lean_runner.py`; the Rust implementation renders candidates and obligations and leaves acceptance to it.

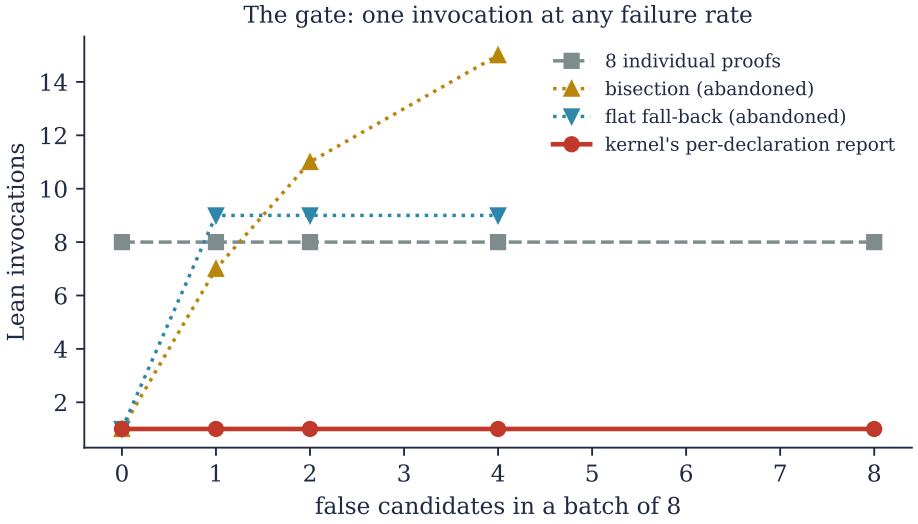


Figure 2: Kernel invocations for a batch of eight as the number of false candidates grows. The two isolation strategies built before the per-declaration report was read are shown for the record; both solved a problem the wrapper had invented.

2.6 The identities, proved

The algebra the dosage machinery and the checkpoint rest on is itself proved in Lean and gated through the same `#print axioms` discipline, in `formal/ANSE/AlixKalSpec.lean` (namespace

`ANSE.Alix`): dosage through + has coefficient 1 (`aurelia_dosage_add`), through the right of − coefficient −1 (`alexis_dosage_sub`), through ∗ the sibling's value (`ilene_dosage_mul`); the backward pass composes multiplicatively (`kal_dosage_chain`); the required value restores the target whenever $B \neq 0$ (`xav_required_value`); and a single disagreement refutes universal equality, so the checkpoint can never refuse a provable candidate (`alix_checkpoint_refutes`, which depends on no axioms at all). Six declarations, zero `sorry`, every axiom on the whitelist.

3 Reference implementation

`crates/kernel_gated_evolution` (Rust, edition 2021) implements the genome, exact evaluation, fitness, variation and selection, one-pass dosage, required values, the inert-locus analysis, the checkpoint, Lean rendering and the search loop, with `num-rational`/`num-bigint` for arbitrary precision and `rand_pcg` for replayable seeds. Six tests, all in exact arithmetic on random genomes, assert the invariants above. The `kge ladder` binary runs the calibration ladder, reports solve rates, distinct solutions and the inert fraction of each run's best individual, exercises the checkpoint as a control with a known answer (each rung's target survives its own checkpoint and is refuted by the other five: exactly 1 passed and 5 refuted), and writes Lean candidate files with a `RungSpec` for the gate. `kge solve` takes any target in prefix notation over the grammar (anything outside $\{+, -, *\}$ is refused, not approximated), runs KGE against random search and a restart hill-climber on the same seeds, points and budget, and renders the solutions for the gate; `kge hillclimb` is the paired control of §4.3; `-points-file` replays a run on exactly another implementation's sample points.

4 Results

4.1 Calibration ladder

Six reachable targets of increasing size, the last being the Verlet position update $q + hp - \frac{1}{2}h^2w^2q$ written in the grammar:

rung	target	size
L1	$q + hp$	5
L2	$q + hp - q$	7
L3	$q + hp - wq$	9
L4	$q + hp - (hw)q$	11
L5	$q + hp - (hw \cdot \frac{1}{2})q$	13
L6	$q + hp - ((hh)(ww) \cdot \frac{1}{2})q$	17

4.2 Solve rates

Two hundred seeds per rung at a budget of 4,000 fitness evaluations, on the crate's own seeded sample points. The last two columns are the kernel's verdicts: how many of the distinct solutions (capped at eight per rung) the Lean kernel proved equal to the specification in one invocation per rung, and whether the deliberately wrong candidate planted in each batch was rejected.

rung	size	solved	rate	median evals	distinct	kernel-proved	control rejected
L1	5	110/200	0.550	1200	82	8/8	yes
L2	7	194/200	0.970	192	63	8/8	yes
L3	9	51/200	0.255	1124	50	8/8	yes
L4	11	4/200	0.020	1206	4	4/4	yes
L5	13	3/200	0.015	1411	3	3/3	yes
L6	17	0/200	0.000	–	0	–	–

Every one of the 31 distinct solutions the search produced was proved by the kernel (31/31) in 5 invocations, and the planted wrong candidate was rejected by name in every rung. On this ladder, every candidate the search believed was right, was. The checkpoint control also came out exactly right at every rung (each target survives its own checkpoint and is refuted by the other five), and the inert fraction of every best individual was zero: at this size cap the search produced no dead code.

Attributing a cross-implementation difference. The Rust solve rate at L3 is lower than the Python prototype's (0.255 against 0.470). The two use different sample-point draws and different random-number generators, so the gap is not attributable by inspection; `kge ladder -points-file` replays a run on exactly the prototype's twelve points. Result: L1: 0.770 on the Python prototype's points against 0.750 measured by the prototype itself and 0.550 on the Rust draw; L3: 0.375 on the Python prototype's points against 0.438 measured by the prototype itself and 0.255 on the Rust draw. Same code, different points: the move from 0.255 to 0.375 at L3 is significant (Fisher exact $p = 0.0131$), while the residual against the prototype's own 0.470 is not ($p = 0.068$) – consistent with random-number-stream variance, though a remaining implementation difference is not excluded at this sample size. A solve rate that moves with the sample points is a property of the fitness-landscape instance, not of the implementation, and the paper's figures are therefore reported with the point set that produced them.

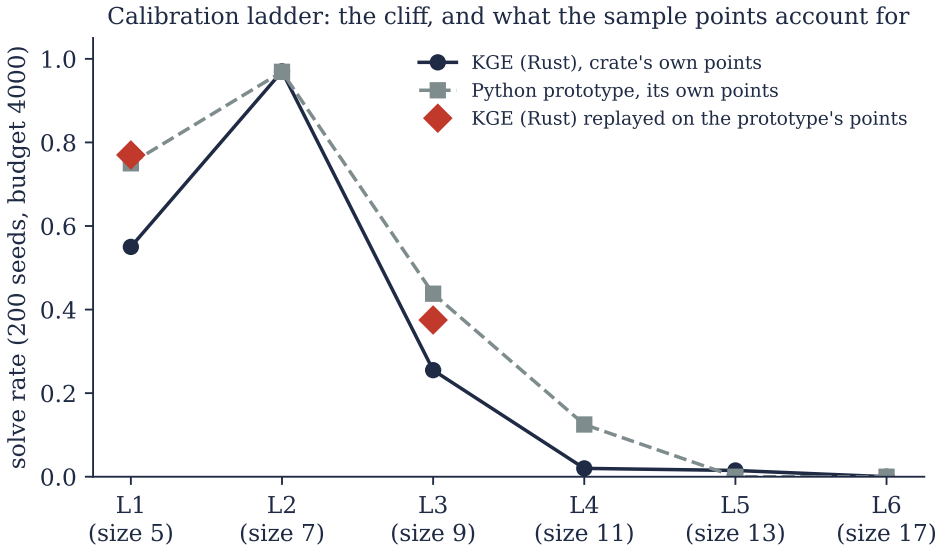


Figure 3: Solve rate by rung for the Rust implementation on its own sample points, the Python prototype on its own, and the Rust implementation replayed on the prototype's exact points. The replay is what attributes the gap.

4.3 Population versus hill-climbing

The simplest thing that could possibly work is a single individual doing greedy descent over sampled mutations, with no population, no selection and no crossover. It was run on the same seeds, sample points and fitness budget as the population, with restarts paid out of that same budget, so neither method receives free work. The paired columns count seeds on which exactly one method solved.

rung	size	population	descent	descent+restarts	GA-only	climb-only	McNemar p
L1	5	0.550	0.005	0.725	31	66	0.00049
L2	7	0.970	0.100	1.000	0	6	0.0312
L3	9	0.255	0.000	0.055	49	9	8.96e-08
L4	11	0.020	0.000	0.010	4	2	0.688
L5	13	0.015	0.000	0.000	3	0	0.25
L6	17	0.000	0.000	0.000	0	0	1

Significant at 0.05 by exact McNemar on the discordant pairs: L1 (the hill-climber with restarts, $p = 0.00049$), L2 (the hill-climber with restarts, $p = 0.0312$), L3 (the population, $p = 8.96e-08$). Single descent without restarts is poor everywhere, so the local optima are real; restarting from fresh random trees is what recovers them, and the population should be read as one way of maintaining diversity among several rather than as the source of the search's power. KGE's guarantees do not depend on which is used: the checkpoint, dosage analysis and kernel gate apply to candidates from either.

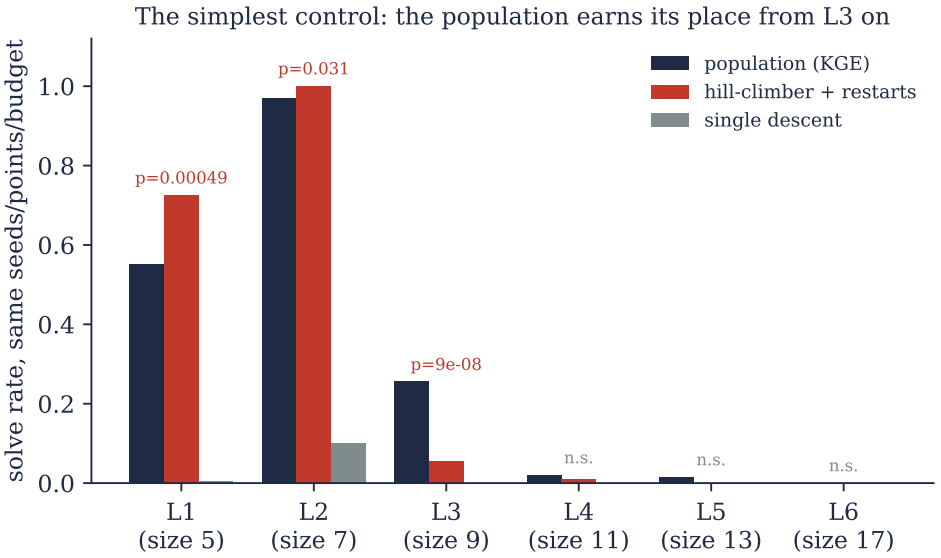


Figure 4: The simplest control. Paired on seeds, points and budget, with restarts paid from the budget; exact McNemar on the discordant pairs. The climber wins the easy rungs; the population wins decisively from L3.

4.4 Landscape around the solutions

Walking away from each rung's exact solution with $k = 1..5$ mutations and comparing against random trees drawn as the initial population is, the basin of useful signal is about four mutations

wide at every rung, neutrality at the solution is 5–8% throughout, and one mutation from a solution still beats a random tree at every rung (ratios 0.61–0.91). The landscape around a solution therefore offers gradient at every size; what grows with target size is the space in which that basin has to be found.

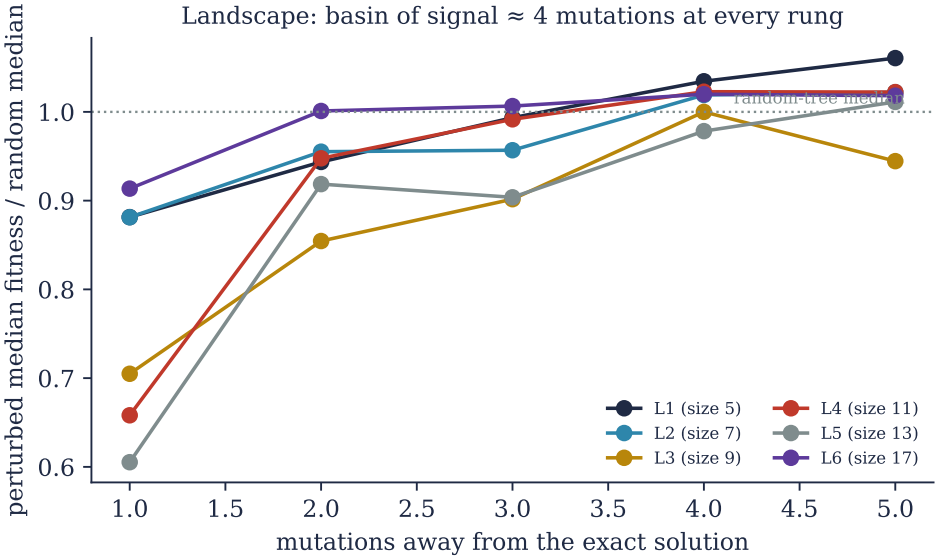


Figure 5: Walking away from each rung's exact solution. The perturbed median relative to a random tree's reaches 1 at about four mutations for every rung, so the local geometry does not distinguish the solvable rungs from the unsolvable ones.

5 Lifetime learning of constants: the Baldwin effect, exactly

In living systems an individual adapts within its lifetime and selection acts on the adapted phenotype — the *Baldwin effect*; in its Lamarckian variant the adaptation is inherited. KGE's dosage machinery gives an exact form of both. For a constant leaf with value v, the required value at each sample point is $v^* = v + (\text{want} - \text{out})/B$ (proved: `xav_required_value`). One leaf holds one value, so it is set only when v^* is the *same* rational at every point where it is not inert. Under *Baldwin* the solved constants determine the fitness but the genome kept in the population is unchanged; under *Lamarck* they are written back. A right-shaped tree with a wrong coefficient becomes a solution in one step; a wrong shape is untouched (both asserted in the crate's tests: L5 with 3 in place of $\frac{1}{2}$ is repaired to L5 itself, and L4's shape against L6's values is left alone).

All three arms used the same seeds, sample points and fitness budget, so each rung is a paired comparison. Because tuning spends dosage passes the fitness budget does not see, total tree-point work is reported alongside, and the plain population was also run at a budget that matches the learning arm's total work.

		solve rate					median tree-point work	
rung	size	plain	Baldwin	Lamarck	B-only / plain-only	McNemar p	plain	Baldwin
L1	5	0.550	0.835	0.835	57 / 0	1.4e-17	41274	29226
L2	7	0.970	1.000	1.000	6 / 0	0.031	2382	2994
L3	9	0.255	0.450	0.450	39 / 0	3.6e-12	48000	108246
L4	11	0.020	0.045	0.045	5 / 0	0.062	48000	132372
L5	13	0.015	0.060	0.060	9 / 0	0.0039	48000	134052
L6	17	0.000	0.000	0.000	0 / 0	1	48000	135216

Across all six rungs and 1,200 paired seeds, lifetime learning lost 0 solves the plain population found: it only ever adds. Baldwin and Lamarck coincided on 0 of 1,200 seeds in outcome and evaluation count — not by accident: a constant is set only when its required value agrees at every sample point, which in practice happens only when setting it solves the tree, so tuning is all-or-nothing and the genome the population keeps never differs between the two arms before a solution is found. All 40 of the 40 distinct solutions gated were proved by the kernel in 5 invocations, with the planted wrong candidate rejected in every batch.

At matched cost. From L3 upward learning spends 2.3–2.8× the tree-point work. The plain population was therefore re-run at a budget of 11,000 fitness evaluations, which matches the learning arm's total work:

rung	plain (4,000)	plain (11,000)	Baldwin (4,000)	Baldwin tree-points
L3	0.255	0.395	0.450	108246
L4	0.020	0.045	0.045	132372
L5	0.015	0.040	0.060	134052
L6	0.000	0.000	0.000	135216

At matched work lifetime learning is ahead on 2 of 4 rungs (L3, L5) and level or behind on 2 (L4, L6). The full Verlet target remains unsolved by every arm at this budget. Lifetime learning is therefore the default the reference implementation should run with where it leads at matched cost, and the comparison above is the evidence for where that is.

6 Population-genetic foundations, proved (H11, H12)

Two classical results bear directly on a mutation-and-selection engine, and both are now proved in Lean over $\mathbb{Q}$ in `formal/ANSE/AlixPopulationGenetics.lean` (eleven declarations in edition 4, sixteen now), zero `sorry`, every axiom on the whitelist, gated through the same `#print axioms` discipline as the rest of the paper. Each is stated at the scope an internal review required, and no wider.

H12: selection never lowers mean fitness — one locus, two alleles, haploid. With allele frequency p, positive constant fitnesses w_1, w_2 and mean fitness $\bar{w} = pw_1 + (1-p)w_2$, one round of selection sends p to $pw_1/\bar{w}$, and

$$\bar{w}' - \bar{w} \;=\; \frac{p(1-p)(w_1 - w_2)^2}{\bar{w}} \;\geq\; 0$$

(`aurelia_mean_fitness_increment`, `aurelia_mean_fitness_nondecreasing`). This is *not* Fisher's general fundamental theorem, which concerns additive genetic variance in arbitrary

systems; it is the one-locus two-allele haploid case, exactly. The numerator is precisely the population's fitness variance, $p(1-p)(w_1-w_2)^2 = p(w_1-\bar{w})^2 + (1-p)(w_2-\bar{w})^2$ (`kal_increment_is_variance`), so the result reads: a round of selection raises mean fitness by $\mathrm{Var}(w)/\bar{w}$. Selection's progress is proportional to the variance present, and stops when it is gone.

H11: i/N is the harmonic solution for Wright–Fisher. For any finite chain on $\{0,\ldots,N\}$ whose interior states each reach state 0 in one step with positive probability, a function with the mean-value property at interior states is determined by its two boundary values (`ilene_harmonic_le_zero`, `ilene_harmonic_unique`): the proof is a maximum principle — at an interior maximum the weighted deviations sum to zero and are all nonnegative, so the weight on state 0 forces the maximum to equal the boundary value. Wright–Fisher satisfies the hypothesis, since from $i < N$ copies the next generation has none with probability $(1-i/N)^N > 0$ (`wfP_pos_zero`); its rows sum to one (`wfP_row`); and j/N has the mean-value property, by the binomial mean (`binomial_mean`, `ilene_wf_harmonic`). Hence any harmonic function with $h(0) = 0$ and $h(N) = 1$ is i/N (`xav_wf_harmonic_solution_unique`). The further step identifying this function with the probability of fixation — an absorption or optional-stopping argument — is *not* formalised here, and we do not claim it.

What they mean for KGE. H12 is the idealised reason a selection-only engine climbs at all, and its variance form gives a measurable prediction we register here and have not yet tested: *runs that stall should be those whose population fitness variance has collapsed*, and an intervention that restores variance (fresh random immigrants, or sample points redrawn so that tied individuals separate) should rescue them at a rate proportional to the variance restored. H11 supplies the neutral baseline for the same question: in a neutral population of size N a single lineage takes over with probability $1/N$ (given the fixation step), so drift alone cannot be counted on to carry a population of 40 across a plateau. Both point at the same lever — maintaining fitness variance — which is the population-level complement of the lifetime learning in §5.

6.1 Toward fixation, and the prediction tested

A finite-horizon fixation bound. The step from "i/N is harmonic" to "i/N is the fixation probability" needs an absorption limit, but half of it needs none. Because j/N has the mean-value property at *every* state, boundaries included (`ilene_wf_harmonic_everywhere`), its expectation is conserved by every step (`alix_expectation_step`, `alix_expectation_iterate`); distributions stay nonnegative (`alix_iterate_nonneg`); and since $j/N \geq 0$ with value 1 at N, *the probability of having fixed by generation t is at most i/N, for every t* (`xav_wf_fixation_by_time_le`). The file now holds sixteen declarations, all gated. The matching lower bound as $t \to \infty$ remains unformalised.

The prediction, tested without the obvious confound. Edition 4 registered, untested, that stalled runs should be those whose population fitness variance has collapsed. Collapse was fixed in advance as *at most two distinct fitness values in the population at a generation boundary.* Comparing how often solved and unsolved runs were collapsed is biased by construction — solved runs stop early, often before a population has had time to collapse — so the test is a per-generation *solve hazard* in each state: a run is uncollapsed until its first collapse and collapsed afterwards (treating collapse as absorbing, which biases against finding a difference), and its solve, if any, is credited to the state it happened in. Under equal rates the split of solves follows a binomial set by the exposures, which gives an exact test.

rung	solves per generation, uncollapsed	solves per generation, collapsed	exact p
L3	53/6,557 = 0.0081	37/7,679 = 0.0048	0.015
L4	5/8,316 = 0.0006	4/10,934 = 0.0004	0.51
L5	11/8,730 = 0.0013	1/10,427 = 0.0001	0.0019
L6	0/6,837 = 0.0000	0/12,963 = 0.0000	1

Where there are enough solves to tell, the prediction holds: after collapse the per-generation chance of solving falls at L3 and L5, and both survive Bonferroni over the three informative rungs. The instrumented runs reproduce the earlier Baldwin solve counts exactly, so measuring did not perturb the search.

The obvious intervention does not work, and H12 says why. On collapse, replacing the worst half of the population with fresh random trees (paid from the same budget) was compared on the same seeds:

rung	Baldwin	+ immigration	immigration-only / Baldwin-only	McNemar p
L3	90	94	10 / 6	0.45
L4	9	9	1 / 1	1
L5	12	12	1 / 1	1
L6	0	0	0 / 0	1

No rung improves. Read through H12 this is coherent rather than contradictory: selection raises mean fitness by the variance *among the individuals it keeps*, and random trees are so much worse than a converged population that they are culled before they contribute. The variance that matters is variance near the current optimum. We register that as the next hypothesis, untested: *restoring variance by perturbing the best individuals, rather than by injecting random ones, should raise the solve hazard of collapsed runs.*

7 Benchmark on Hugging Face datasets

To test the engine outside the ladder it was calibrated on, we took the SRSD-Feynman datasets (`yoshitomo-matsubara/srsd-feynman_{easy, medium,hard}` on the Hugging Face Hub, CC-BY-4.0): 120 physics laws with a symbolic ground truth each. Selection was mechanical and is deposited: an equation is a candidate only if it is a polynomial in at most four variables with coefficients in $\mathbb{Z}[\frac{1}{2}]$, i.e. inside the reference grammar; everything else is refused with its reason. The equation strings are untrusted data and were parsed by walking a syntax tree, never evaluated.

8 of 120 laws fit (6.7%). The five selected, smallest first and one per algebraic shape, are I.12.1 ($F = \mu N$), I.39.10 ($E = \frac{3}{2}pV$), I.14.4 ($U = \frac{1}{2}kx^2$), II.34.2 ($\mu = \frac{1}{2}qvr$) and II.37.1 ($E = \mu B(1 + \chi)$). Each was run with `kge solve` against random search (the naive floor) and the restart hill-climber (the simple control) on the same hundred seeds, twelve sample points and budget of 4,000 evaluations, and against the traditional method for a polynomial target — exact regression over the 35 monomials of degree ≤ 3 in four variables, solved by Gaussian elimination over the rationals — in two regimes: on the same twelve points, and on forty.

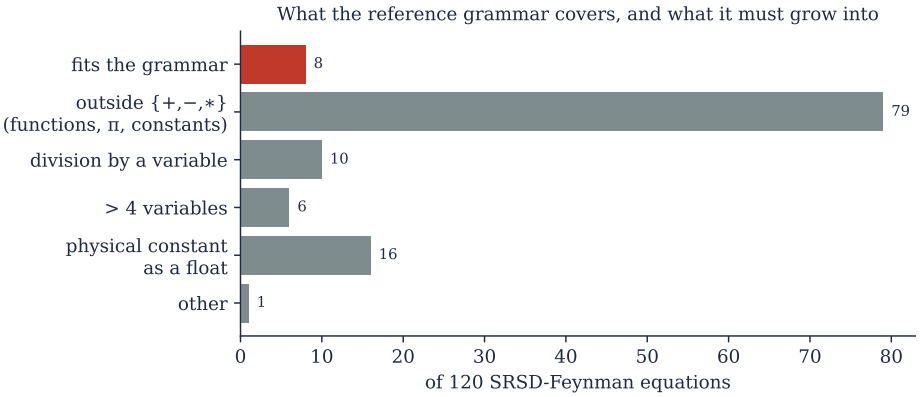


Figure 6: Grammar coverage of SRSD-Feynman. Eight of 120 laws lie inside $\{+, -, *\}$ with dyadic coefficients and at most four variables; the refusal reasons are the roadmap for the grammar.

			KGE		climber	random	exact regression	
law	form	size	rate	med. evals	rate	rate	12 points	40 points
I.12.1	$x0\,x1$	2	1.00	184	1.00	1.00	ambiguous (null. 23)	exact
I.39.10	$1.5\,x0\,x1$	3	0.51	1180	0.21	0.10	ambiguous (null. 23)	exact
I.14.4	$0.5\,x0\,x1^2$	4	0.07	525	0.27	0.20	ambiguous (null. 23)	exact
II.34.2	$x0\,x1\,x2/2$	4	0.22	1135	0.31	0.31	ambiguous (null. 23)	exact
II.37.1	$x0\,x2\,(x1+1)$	4	0.10	1102	0.16	0.15	ambiguous (null. 23)	exact

Across the five laws KGE solved 190 of 500 runs, the restart hill-climber 195 and random search 176. Every KGE solution was rendered to Lean and proved against the law by the kernel (§7.1). The traditional method is the honest foil: on the twelve points the search solves from, dense regression over 35 monomials is underdetermined for every law (nullity 23) and its particular solution does not reproduce the law on held-out points; given forty points it recovers each law exactly in about half a second with one or two nonzero terms. Which method wins depends on how much data there is: with few exact samples the search for a small expression recovers what a dense fit cannot pin down, and with enough samples linear algebra is faster and certain. KGE's distinctive contribution is not the recovery but what follows it — a kernel proof that the recovered form equals the law. The population wins outright on 1 of the five laws (I.39.10) and is beaten by at least one of the two naive baselines on 3 (I.14.4, II.34.2, II.37.1). Every Hub law here has at most four leaves, which is the regime where the ladder already showed the population to be overhead (§4.3): for tiny targets, random search over small trees and a restart hill-climber are the better search layer, and the guideline in §9 says so. The kernel gate is indifferent to which layer found the candidate.

7.1 Kernel verdicts on the Hub solutions

Every distinct KGE solution (capped at eight per law) was rendered as a Lean definition with a theorem of equality to the law, and the batch for each law was compiled in one invocation with a deliberately wrong candidate (the law plus one) appended.

law	gated	proved	rejected	control rejected	invocations	wall clock
I.12.1	8	8	0	yes	1	6.5 s
I.39.10	8	8	0	yes	1	6.3 s
I.14.4	7	7	0	yes	1	6.2 s
II.34.2	8	8	0	yes	1	6.3 s
II.37.1	8	8	0	yes	1	6.3 s

39 of 39 distinct solutions proved, in 5 invocations; the planted wrong candidate was rejected in every batch .

8 Where KGE applies

Any synthesis with a formal acceptance criterion. The architecture is indifferent to the domain provided three things exist: an exact evaluator for candidates, a renderable specification, and a kernel that can decide equality. The search layer can be replaced wholesale; the guarantee is in the gate. Concretely: rewriting rules in a compiler whose equivalence must be certified; algebraic simplifications in a computer-algebra pipeline; closed-form identities conjectured from data and then proved rather than tested.

Lean-gated numerical kernels. The motivating case: evolving a candidate kernel (an integrator step, a stencil, an update rule, a finite-difference coefficient set) that must be *proved* equal to a specification before it enters a simulation. KGE finds candidates by exact fitness, removes most wrong ones at zero kernel cost, and proves the survivors in batches. The Verlet position update is the worked example; the same shape covers symplectic integrators, conservation-law discretisations and any update rule expressible over the grammar.

Inert-locus analysis of evolved programs. Inert loci are identified exactly, not statistically, from the dosage coefficients. Evolved programs are notorious for bloat; KGE measures it as a fraction of loci that cannot reach the phenotype and can prune it with no change in behaviour. The same analysis gives an exact sensitivity map of any program over the grammar: which inputs reach the output, and with what coefficient, at every point.

eXpensive verification, batched. The one-invocation gate is independent of KGE: any pipeline that verifies candidates in Lean — an LLM-driven prover loop, a conjecture miner, a refactoring assistant — can read the per-declaration report and verify a batch at the price of one, at any failure rate, with every culprit named.

Rediscovery of laws from exact data. Where a physical law is polynomial in its variables with small rational coefficients (§7 measures how often that holds), KGE recovers it from a dozen exact samples and proves the recovered form, where dense regression on the same samples is underdetermined. The complement — laws with constants, roots and transcendental functions — is the grammar's roadmap, not its claim.

9 Guidelines: where and how to use KGE

When to reach for it. Use KGE when a candidate's acceptance must be *formal* rather than empirical, when an exact evaluator exists for the candidate representation, and when the

specification can be rendered as a Lean definition. If any of the three is missing, the kernel gate is the piece to keep and the search layer the piece to swap.

eXact first. Keep every quantity the search sees exact. The checkpoint's soundness and the dosage identities both rest on exact arithmetic; a floating-point fitness would turn refutations into guesses.

Arbiter before operator. Build and test the kernel gate with a planted wrong candidate in every batch *before* tuning any variation operator. A gate that cannot reject is worse than none, and no operator result means anything until the gate is shown to refuse.

Verify against the simplest control. Run the restart hill-climber on the same seeds, points and budget before claiming anything for the population, and report McNemar on the discordant pairs. On easy targets the climber wins (§4.3); design the population in because the hard rungs need it, not because the method is called genetic.

Calibrate difficulty on a pooled band. Choose benchmark targets by a pooled success-rate band fixed in advance, so that neither trivial nor impossible targets masquerade as a comparison, and never by the rung that flatters a method.

Report both costs. Fitness evaluations and total tree-point evaluations, always. An operator that probes the genome spends compute the budget does not see; a win in one currency and a loss in the other is a loss.

Replay, do not inspect. When two implementations disagree, replay one on the other's exact sample points. A solve rate that moves with the points is a property of the landscape instance, not a bug.

Grow the grammar deliberately. Every refusal in §7 is a feature request with a count attached: division by a variable, transcendental functions, physical constants. Add them with their dosage rule proved, as the three operators here are, or not at all.

10 Significance

For science. The step that scientific computing most needs and least has is a mechanical bridge from a candidate formula to a certified one. KGE makes that bridge the *default* output of a search: what comes out is not a formula that fits, but a formula with a kernel-checked proof that it equals the specification on the whole domain. For laws inside the grammar, that is rediscovery with a proof attached, from a dozen exact samples.

For verified software. Numerical kernels in simulators, control systems and scientific libraries are routinely hand-derived and empirically tested. A pipeline that evolves them and refuses to accept without a proof moves the guarantee from the test suite to the kernel, and does so at one invocation per batch of candidates.

For the collaboration between people and machines. The division of labour KGE enforces — the search may be heuristic, biological, even wrong, because the kernel decides — is a model for working with AI systems in general: let the generator be creative and the arbiter be formal, and never let the generator grade itself. Every control in this paper exists because that principle was applied to our own work as well as to the machine's.

11 Limitations

- **Scope of the reference implementation.** One closed grammar $\{+, -, *\}$, four variables, three constants, one specification family. The architecture is general; the measurements are not.
- **Scaling.** Solve rates fall steeply above targets of about nine nodes, and the full Verlet target (size 17) is not solved within 4,000 evaluations. The cliff is not explained by the geometry of the basin around the solution (measured) and is not relieved by recombination: eight variation operators including homologous, contribution-preserving and semantically back-propagated variants were tested against mutation under pre-registered bars and none improved on it, which is why KGE is mutation-driven. The evidence points at how hard the basin is to find in a growing space, not at how it is shaped. (Companion measurements: Paper I, DOI 10.5281/zenodo.23622868.)
- **The checkpoint is one-sided by design.** A candidate can be tuned to agree at the sample points and still be wrong; it survives the checkpoint and costs a kernel invocation before being rejected. This is a cost, never a soundness risk, and is reduced by drawing sample points per generation.
- **Dosage is instrumentation.** As a guide for variation operators it did not improve search outcomes in our tests; its value here is exact analysis and exact pruning, not steering.
- **Fixed per-invocation cost.** About 97% of a Lean invocation is fixed overhead on this host. Batching amortises it; a persistent kernel process would remove it and was not built.

12 Future research

1. **Keep exploiting the affine structure.** Because the output is affine in every locus, constant loci could be solved by exact linear algebra rather than searched; a genome of structural operators with coefficients fitted exactly would turn part of the search into a solve — and §7 shows exactly where dense regression needs more data than the search does, so the hybrid has a measured motivation.
2. **Adaptive checkpoints.** Sample points redrawn each generation defeat evaders at no proof cost and bound how long a wrong survivor can persist.
3. **Lean kernel, persistent.** Removing the fixed per-invocation cost would make the gate nearly free per candidate and change what batch sizes are optimal.
4. **Grammar growth with proofs.** Division, powers and the common transcendental functions, each added with its dosage rule proved in `AlixKalSpec.lean`; the SRSD-Feynman refusals in Fig. 6 rank them by how many laws each would unlock.
5. **Representations with higher solution density at scale.** The cliff is about finding basins in a growing space; semantic deduplication of the population is the natural lever once the grammar is richer.
6. **Contribution-preserving transfer, properly powered.** Rescaling a transplanted subtree so its dosage-weighted contribution is preserved exactly showed a positive signal against its matched control ($p = 0.031$, not surviving correction for five comparisons) and merits a dedicated study.
7. **Application to the project's Python-genome kernel path.** The AlphaEvolve-style path evolves Python kernels; KGE's checkpoint and gate apply unchanged once a Lean specification exists for the kernel being evolved.

13 Reproduction

The Rust crate, its tests, the ladder runs and the gate results are deposited alongside this paper, together with the Python prototype in which the identities were first verified and the verifier whose per-declaration report makes the one-invocation gate possible. Every number above is produced by a committed script or test and is checked against its artefact before deposit.